

ADENTAN MUNICIPAL EDUCATION OFFICE

GovPay Desk

Individual contribution

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CSIT622 Capstone Project · Group 7

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Augustine Chiful: AI integration and automated checks

I own the emerging-technology component: the Claude API integration and the deterministic checks that run alongside it.

The division between them is the point. Drafting a memo and summarising a voucher for review are language tasks, and a language model does them well. Detecting a duplicate payment is not; it is a query. Spotting an unusual amount is a standard deviation. Matching a description to a budget line is keyword scoring. I implemented those three as arithmetic in `VoucherChecks` because each can then explain itself in one sentence, which matters when the output is shown to someone deciding whether to release public money.

Every AI call is optional and outside the critical path. If the key is missing or the service is slow, the feature is absent rather than broken, and no approval or payment depends on it.

The code this covers

- `app/Services/VoucherIntelligence.php`
- `app/Services/VoucherChecks.php`
- `tests/Feature/VoucherChecksTest.php`

Questions to be ready for

The examiner may ask about any part of the system, not only this one. These are the ones closest to the work described above:

- Why are the three checks arithmetic rather than machine learning?
- What happens to the application if the Claude API is unavailable?
- How does the outlier check work, and why does it need five prior vouchers?

Before submitting, confirm this describes what you actually did. This section is assessed individually, and you may be asked to demonstrate any claim in it. Rewrite anything that does not match your own work, and add what is missing, particularly what you found difficult and what you would do differently now.